



Transferable adversarial examples based on global smooth perturbations

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ABSTRACT

Although the attack rate and the imperceptibility of perturbations are two main concerns of adversarial attacks, the transferability of adversarial examples is an emerging topic due to the need for applications. It is known from previous work that adversarial examples with low-frequency perturbations have better transferability than those with high-frequency perturbations. In this paper, we propose a method to generate global smooth low-frequency perturbations with parameterized smooth functions, unlike previous pixel-wise local methods. We optimize perturbations by minimizing a proposed loss function that fulfills the attack task and meets the requirement of imperceptibility for perturbations. The global smoothness of perturbations ensures the spectrum of low-frequency and hence increases adversarial examples' transferability. In the implementation, the Gaussian mixture model is used as the prototype of parameterized smooth functions to evaluate the proposed method. Extensive experiments on ImageNet and CIFAR-10 demonstrate that our method significantly improves the transferability by 10–20% over other state-of-the-art methods with a comparable attack rate.

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1. Introduction

Deep Neural Networks (DNNs) are well known to be vulnerable to *adversarial attacks* because adding carefully-designed small perturbations to images can lead to misclassifications (Goodfellow et al., 2015). Adversarial attacks reveal the weaknesses of DNNs, even though DNNs have achieved human-competitive performances in many tasks (Krizhevsky et al., 2017; Yang et al., 2020; Zhen et al., 2020). More importantly, adversarial attacks also pose potential security threats to machine learning systems in practice and may stunt the growth of applying DNNs in real applications. Therefore, the study of adversarial attacks is crucial for improving the robustness of DNNs (Madry et al., 2018).

There are three important concerns when designing adversarial attacks. First, the adversarial perturbation should be *imperceptible* to human eyes. It means that an adversarial example is supposed to be similar to the original image so that humans will not detect the difference (Goodfellow et al., 2015; Kurakin et al., 2017; Moosavi-Dezfooli et al., 2016). Second, adversarial examples are expected to have a satisfactory *attack rate* against the attacked DNN, i.e., the target model, to ensure the attack's effectiveness. How-

ever, in practice, many target models are not accessible so that attack methods cannot utilize their information to conduct attacks. One workaround is to generate adversarial examples by attacking some accessible models, i.e., the source models, then transfer the adversarial examples against the target model. Therefore, the third concern is *transferability*, which measures the success rate of source model-based adversarial examples in misleading the target model (Papernot et al., 2017).

Adversarial examples generated by early attacks have achieved high attack rates and keep good similarity with original images when attackers have full knowledge of the architecture and parameters of the target model (Carlini and Wagner, 2017; Kurakin et al., 2017; Moosavi-Dezfooli et al., 2016). However, these adversarial examples cannot transfer well to unseen models because the perturbations overfit the source model (Inkawhich et al., 2020). The low transferability renders these methods less useful when attacking unknown models. Therefore, improving the transferability of adversarial examples comes urgent for adversarial attacks.

Some clues have shown that low-frequency perturbations may prevent adversarial perturbations from overfitting the source model and hence improve the transferability (Lv et al., 2021; Sharma et al., 2019). Low-frequency perturbations represent perturbations that have a low frequency in the frequency domain. Due to the clues, SmoothFool (Dabouei et al., 2020) tried to generate low-frequency perturbations by smoothing high-frequency parts of

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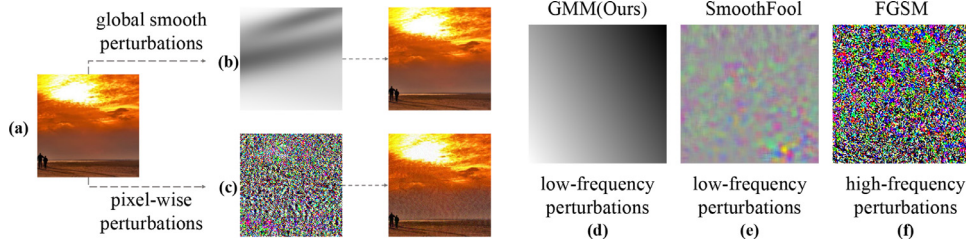


Fig. 1. (a): An original image. (b): The proposed global smooth perturbations and the perturbed image. (c): Previous pixel-wise perturbations and the perturbed image. (d): Low-frequency perturbations generated by GMM (our method). (e): Low-frequency perturbations generated by SmoothFool. (f): High-frequency perturbations generated by other attacks.

pixel-wise perturbations (*pixel-wise* perturbations mean that perturbations added to different pixels are independent, see Fig. 1 (c)). However, perturbations generated by SmoothFool are local smooth (see Fig. 1 (e)) which do not help much in improving the transferability (see Section 4).

In this paper, we propose a method that also aims to design low-frequency perturbations (see Fig. 1). But different from SmoothFool, we generate low-frequency perturbations from parameterized smooth functions f . On one hand, f can generate global smooth perturbations instead of local smooth perturbations. On the other hand, the smoothness of f can ensure the spectrum of the perturbation concentrate on the low-frequency part. Moreover, the number of parameters in f can make a trade-off between adversarial examples' attack rate and transferability (details in Section 4.3). We optimize the parameters of f by minimizing a designed loss function, which considers both attack rate and imperceptibility of perturbations. Specifically, we use a popular metric LPIPS (Zhang et al., 2018) for Image Quality Assessment (IQA) to measure the imperceptibility of perturbations and integrate it into the loss design.

To evaluate our method, we use the Gaussian mixture model (GMM) as the parameterized smooth function, because GMM can approximate any continuous function in the sense of uniform convergence (Girosi and Poggio, 1990). Based on GMM, we conduct extensive experiments on two datasets (ImageNet and CIFAR-10) and compare our method with seven attack methods. The results show that our method achieves 10–20% higher transferability than previous works. Meanwhile, our method maintains comparable attack rates and perturbations' imperceptibility with compared methods. The above two points demonstrate the effectiveness of our attack method.

To sum up, our method can balance the imperceptibility of perturbations, the attack rate, and the transferability of adversarial examples. Our main contributions are as follows:

- To the best of our knowledge, this is the first work applying parameterized smooth functions to generate global smooth perturbations for adversarial attack;
- To ensure the imperceptibility of perturbations, the IQA metric is introduced in the design of loss function;
- The transferability of adversarial examples is significantly improved which is demonstrated by extensive experiments on ImageNet and CIFAR-10 (see Section 4).

2. Related work

Early works in the field of adversarial attacks pay much attention on the attack rate and the imperceptibility of perturbations. Attack methods generate adversarial examples by adding pixel-wise perturbations to the original image I to achieve high attack rates. Pixel-wise perturbations mean that perturbations at different positions are independent of each other. To make the adversarial perturbations invisible, attackers take ℓ_p norm of the pertur-

bation as the constraint condition. For instance, ℓ_0 norm is used in one-pixel attack (Su et al., 2019), ℓ_2 norm in DeepFool (Moosavi-Dezfooli et al., 2016) and ℓ_∞ norm in I-FGSM (Goodfellow et al., 2015). These methods achieve high attack rates when the target model is accessible. However, such adversarial examples are hard to transfer to inaccessible models. Therefore, it is important to improve the transferability of adversarial examples.

Transfer-based methods can be roughly divided in two classes: one improves the transferability on the basis of pixel-wise attacks, the other proposes to design non-pixel-wise perturbations to improve the transferability. For the first class, some transfer-based methods are proposed with pixel-wise perturbations from many aspects. From the gradient aspect, MI-FGSM (Dong et al., 2018) use the momentum gradient, NI-FGSM (Lin et al., 2020) use the Nesterov gradient, TI-FGSM (Dong et al., 2019) and TAIG (Huang and Kong, 2022) use smoothed gradients update perturbations to improve the transferability. From the diversity aspect, DI-FGSM (Xie et al., 2019) and ATTA (Wu et al., 2021) utilize different transformations of input images and these methods can be combined with MI-FGSM or NI-FGSM. From the loss aspect, Wang et al. (2021a) penalize interactions during the attacking process and Zhu et al. (2022) propose AAI as an effective metric for transferability. The above works generate perturbations in the output layers. Besides, some works (Inkawich et al., 2020; 2019; Nakka1 and Salzmann, 2021; Wang et al., 2021b; Yang et al., 2021) believe that outputs are model-specific but intermediate layer features are task-specific. Therefore, they apply attacking in the intermediate layer features rather than outputs.

Besides focusing on pixel-wise perturbations, the other class of transfer-based methods focuses on improving the transferability by designing non-pixel-wise perturbations. Non-pixel-wise perturbations mean that the perturbations on adjacent pixels are correlated. ColorFool (Shamsabadi et al., 2020) segments the original images with several masks and perturbed different channels for different masks. ColorAdv (Laidlaw and Feizi, 2019) applies linear interpolation functions on the original images in the color space and enforces similar colors to be perturbed similarly. Observing that adversarial examples with low-frequency perturbations transfer better than those with high-frequency perturbations (Sharma et al., 2019), SmoothFool (Dabouei et al., 2020) generates perturbations by smoothing high-frequency perturbations with smooth kernels.

SmoothFool is the most similar work to this paper. The main differences between SmoothFool and this work are: (i) Perturbations are generated in different ways. We generate perturbations with parameterized smooth functions while SmoothFool smooths perturbations from DeepFool; (ii) Properties of perturbations are different. SmoothFool generates local smooth perturbations, but we generate global smooth perturbations with parameterized smooth functions.

In addition to the works introduced above, some works study the transferability of adversarial examples in visual transform-

ers (Naseer et al., 2022), automatic speech recognition systems (Abdullah et al., 2022), transfer learning (Liang et al., 2021) and so on, which is beyond the scope of this paper.

3. Methodology

We will describe how to generate perturbations with parameterized smooth functions f in Section 3.1. The initialization of the parameters in f and the loss function design are introduced in Sections 3.2 and 3.3, respectively. Section 3.4 will describe how to update the parameters in f . In Section 3.5, we prove that perturbations generated by our method are low-frequency through statistical analyses. Algorithm 1 gives a pipeline of our method.

Algorithm 1 Global Smooth Perturbations

Require: Source model M_s ; Image I with ground truth class label t ; Normalized pixel coordinates (x, y) ; Adversarial loss $\mathcal{L}_{adv}$, perceptual loss $\mathcal{L}_{dis}$ and balancing weight λ ; Perturbation scale ρ ; Constraint ε for $\mathcal{L}_{dis}$.

Ensure: An adversarial example I' .

- 1: Choose a parameterized continuous function $f(x, y; \Theta)$
 - 2: Initialize Θ with the JND mask (details in Section 3.2)
 - 3: Initialize $I' = I$
 - 4: **for** iter = 1 **to** Max_iteration **do**
 - 5: The perturbation $\Delta I(x, y) = f(x, y; \Theta)$, $\Delta I_{min}/\Delta I_{max}$ are the minimum/maximum values in ΔI respectively
 - 6: Normalization $\Delta I = (\Delta I - \Delta I_{min})/(\Delta I_{max} - \Delta I_{min})$
 - 7: $I' = \text{clip}_{[0,255]}[I + \rho \cdot \Delta I]$
 - 8: $\mathcal{L} = \mathcal{L}_{adv} + \lambda \cdot \mathcal{L}_{dis}$
 - 9: Update Θ by $\nabla_{\Theta} \mathcal{L}(I', t)$ with the Adam optimizer
 - 10: **if** $M_s(I') \neq t$ and $\mathcal{L}_{dis} < \varepsilon$ **then**
 - 11: **Return** I'
 - 12: **end if**
 - 13: **end for**
-

3.1. Perturbations generation with parameterized smooth functions

Let M_s be the source model, I be an input image, and $M_s(I)$ be the prediction result of I by M_s . The range of pixel coordinates of I is normalized to $\Omega = [-1, 1] \times [-1, 1]$.

The perturbation ΔI is generated by parameterized functions of the following form $\Delta I(x, y) = f(x, y; \Theta)$ where Θ denotes the parameters for f , (x, y) is the position coordinate of the pixel in Ω . The same function f is applied to all positions of I . The perturbation range is normalized to $[0, 1]$. Then, ΔI is added to the input image I with a strength control ρ to generate the perturbed image I' , i.e.

$$I' = \text{clip}_{[0,255]}[I + \rho \cdot \Delta I] \quad (1)$$

where $\text{clip}_{[0,255]}$ denotes clipping the intensity of pixels in I' into $[0, 255]$. The same perturbation is applied to RGB channels to avoid potential color distortion. Due to the smoothness and approximation property of the Gaussian mixture model (GMM) (Girosi and Poggio, 1990), GMM is used as the parameterized function. The perturbation added to the pixel at location (x, y) is

$$\Delta I(x, y) = f(x, y; \Theta) = \sum_{i=1}^K p_i \cdot G(x, y; \mu_i, \Sigma_i) \quad (2)$$

where p_i ($i = 1, \dots, K$) are weights of K Gaussian bases, and each $G(x, y; \mu_i, \Sigma_i)$ is equal to

$$\frac{1}{2\pi\sigma_{x,i}\sigma_{y,i}} e^{-\frac{1}{2}\left[\left(\frac{x-\mu_{x,i}}{\sigma_{x,i}}\right)^2 + \left(\frac{y-\mu_{y,i}}{\sigma_{y,i}}\right)^2\right]} \quad (3)$$

with mean vector $\mu_i = (\mu_{x,i}, \mu_{y,i})^T$ and covariance matrix $\Sigma_i = \text{diag}(\sigma_{x,i}, \sigma_{y,i})$ ($i = 1, \dots, K$).

3.2. Parameters initialization

To generate unnoticeable perturbations, we take the influence of image content into account when initializing the parameters in $f(x, y; \Theta)$. Because some Just-Noticeable-Difference (JND) works show that JND, a metric reflecting the visibility of perturbations, is related to the image content (Yu and Winkler, 2013). In detail, JND is a threshold below which human eyes cannot sense any changes, and positions with large JND values can tolerate large perturbations (Liu et al., 2010). We use the algorithm from the work (Liu et al., 2010), which utilizes the image content, to calculate the JND value for each pixel. According to the Eq. (3), the magnitude of perturbations is largest at $(\mu_{x,i}, \mu_{y,i})$ ($i = 1, \dots, K$) in each Gaussian basis. Therefore, the mean vector μ_i ($i = 1, \dots, K$) of K Gaussian bases are initialized at the positions with top K JND values. Moreover, we initialize $\Sigma_i = \text{diag}(1, 1)$ ($i = 1, \dots, K$).

3.3. Loss design

We design the loss function to accommodate both attack success and the imperceptibility of perturbations. The loss function is a weighted sum of two parts: the adversarial loss $\mathcal{L}_{adv}$ which accounts for misleading the source model M_s , and the perceptual loss $\mathcal{L}_{dis}$ which measures the perceptual distance between I' and I .

The used $\mathcal{L}_{adv}$ was proposed in CW attack (Carlini and Wagner, 2017), i.e.,

$$\mathcal{L}_{adv}(I, M_s) = \max_{l \neq t} \left(\max\{M_s(I')[l]\} - M_s(I')[t], \tau \right) \quad (4)$$

where $M_s(I')[l]$ represents the l th element of the logit vector, t is the true label of the image and τ is a hyper-parameter.

As for $\mathcal{L}_{dis}$, previous works used $\|\Delta I\|_p$ as their $\mathcal{L}_{dis}$ (Carlini and Wagner, 2017; Goodfellow et al., 2015). However, it is well known that the ℓ_p norm does not correlate well with the human vision system from the study of Image Quality Assessment (IQA) (Wang and Bovik, 2009). In the field of IQA, there are some metrics that fit human vision well (Prashnani et al., 2018; Wang et al., 2004; Zhang et al., 2018). Therefore, we pick a popular IQA metric, LPIPS (Zhang et al., 2018), as $\mathcal{L}_{dis}$ in this paper. LPIPS measures the perceptual distance between two images and the smaller LPIPS is, the more similar two images are. To conclude, $\mathcal{L}_{dis}(I, I') = \text{LPIPS}(I, I')$ in this work.

We think an adaptive trade-off between $\mathcal{L}_{adv}$ and $\mathcal{L}_{dis}$ instead of a fixed weight is more appropriate. First, when ΔI is small, $\mathcal{L}_{adv}$ is appropriate to have a large weight. Because human can not distinguish between I' and I when ΔI is small (Liu et al., 2010; Qi et al., 2016; Yu and Winkler, 2013). Second, when ΔI makes I' obviously different from I , $\mathcal{L}_{dis}$ needs to have a large weight. Because adversarial examples are expected to be similar to original images (Goodfellow et al., 2015). According to the above two reasons, we design the adaptive loss function as follows:

$$\mathcal{L} = \begin{cases} \mathcal{L}_{adv} & |f > J_I| < \phi, \\ \mathcal{L}_{adv} + \lambda \cdot \mathcal{L}_{dis} & \text{otherwise} \end{cases} \quad (5)$$

where $|f > J_I|$ represents the proportion of pixels whose perturbation value $f(x, y; \Theta)$ is larger than the JND value $J_I(x, y)$, ϕ is a hyper-parameter and λ is a balance weight.

3.4. Parameters update and stopping criterion

The parameters in $f(x, y; \Theta)$ are optimized by the Adam optimizer (Kingma and Ba, 2015) to minimize the designed loss. Previous works stop their algorithms either when $M_s(I') \neq t$ (Moosavi-Dezfooli et al., 2016) or when the iteration reaches the predefined maximum iteration number (Dong et al., 2018; Lin et al.,

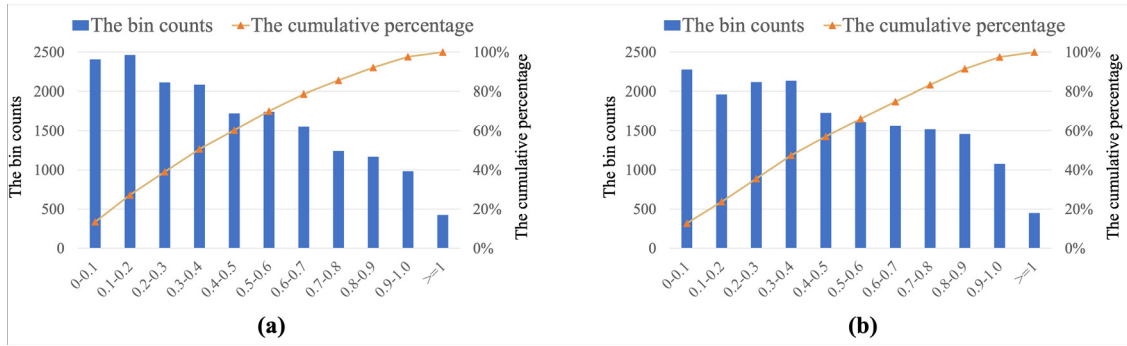


Fig. 2. The distribution of (a) $|\mu_{x,i}|$; (b) $|\mu_{y,i}|$ of 20 Gaussian bases over 1000 perturbations against ResNet50. The horizontal axis represents the interval of the data, blue histograms represent the bin counts in each interval and the orange line represents the cumulative percentage. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2020). Different from them, we add $\mathcal{L}_{\text{dis}} < \varepsilon$ as a stopping criterion of our algorithm to guarantee that I' is similar to I (Line 10 in Algorithm 1).

3.5. The properties of low-Frequency

A toy experiment is conducted to show that GMM indeed generates low-frequency perturbations. We randomly choose 1000 images from ImageNet (Deng et al., 2009), use GMM with $K = 20$ in Eq. (2) to attack the widely-used baseline ResNet50 (He et al., 2016) and calculate distributions of mean vectors of 20 Gaussian bases over all perturbations. The distributions are shown in Fig. 2.

Mean vectors are concerned as they determine the frequency of Gaussian bases by the Fourier transform of Eq. (3)

$$F_i(u, v) = e^{-2\pi^2((u\sigma_{x,i})^2 + (v\sigma_{y,i})^2)} - 2\pi i(u\mu_{x,i} + v\mu_{y,i}). \quad (6)$$

Eq. (6) implies that $\mu_{x,i}$, $\mu_{y,i}$ with small absolute values can produce low-frequency perturbations. As the range of the mean vectors is $[-1, 1] \times [-1, 1]$, we assume that $\mu_{x,i}$, $\mu_{y,i}$ within $[-0.5, 0.5]$ have small absolute values. Fig. 2(a) and (b) show the distribution of $|\mu_{x,i}|$, $|\mu_{y,i}|$ ($i = 1, \dots, 20$). The one-proportion z-test (Weiss, 2016, p.558) is conducted to explore, at the 5% significance level, whether a majority of $\mu_{x,i}$ and $\mu_{y,i}$ have small absolute values. The P-value is 0.039 for the hypothesis test of $\mu_{x,i}$ and 0 for the hypothesis test of $\mu_{y,i}$ (details in the supplementary material). Because the P-value is less than the specified significance level of 0.05, we conclude that a majority of $\mu_{x,i}$ and $\mu_{y,i}$ have small absolute values so perturbations generated by GMM are generically of low-frequency. More results approved by z-tests with the similar conclusion can be found in the supplementary material.

4. Experiment

We conduct experiments on two datasets with nine models and introduce the experimental setup in Section 4.1. We compare our method with seven attack methods and report three evaluation metrics in Section 4.2 under both single source model setup and ensemble source models setup. The ablation studies are summarized in Section 4.3.

4.1. Setup

DataSets. We conduct experiments on two commonly-used datasets: ImageNet (Deng et al., 2009) and CIFAR-10 (Krizhevsky and Hinton, 2009). For ImageNet, we randomly choose 10 sets of images from the ILSVRC 2012 validation set. Each set has 1000 images corresponding to 1000 classes. For CIFAR-10

dataset, the whole test set (including 10,000 images) is divided equally into 10 sets for the experiment.

Target Models. ResNet50, ResNet18 (He et al., 2016), VGG16 (Simonyan and Zisserman, 2015), DenseNet121 (Huang et al., 2017), GoogleNet (Szegedy et al., 2015), MobileNetV2 (Sandler et al., 2018), InceptionV3 (Szegedy et al., 2016), SqueezeNet (Iandola et al., 2016) and AlexNet (Krizhevsky et al., 2017) are nine target models we use¹

Methods under Comparison. GMM is compared with seven attack methods, including three non-transfer-based attacks: DeepFool (Moosavi-Dezfooli et al., 2016), SparseFool (Modas et al., 2019), I-FGSM (Kurakin et al., 2017), and four transfer-based attacks: MI-FGSM (Dong et al., 2018), NI-FGSM (Lin et al., 2020), ColorAdv (Laidlaw and Feizi, 2019), and SmoothFool (Dabouei et al., 2020).

Evaluation Metrics. Methods are compared from three aspects:

(1) *Attack rate*, which represents the proportion of adversarial examples that are misclassified by the targeted model;

(2) *Transferability* from a set of source models $M_s = \{M_{s_1}, M_{s_2}, \dots, M_{s_N}\}$ to the target model M_t is calculated by $|A_{M_t}|/|A_{M_s}|$ where $A_{M_s} = \{\text{examples that attack all models in } M_s \text{ successfully}\}$, $A_{M_t} = \{\text{adversarial examples } \in A_{M_s} \text{ and can attack } M_t \text{ successfully}\}$ and $|\cdot|$ represents the number of elements in the set;

(3) *Imperceptibility* of perturbations, which is evaluated by three popular IQA metrics, SSIM (Wang et al., 2004), LPIPS (Zhang et al., 2018) and PieAPP (Prashnani et al., 2018).

Implementation Details. The perturbation scale ρ in Eq. (1) is 12 because perturbations keep invisible in this case (Dong et al., 2018). The comparison of different attacks is conducted under the same perturbation strength where $\ell_\infty = 12$ and other settings are the same as recommended in the original papers. The number K of Gaussian bases in Eq. (2) is 20, chosen according to the ablation study (Section 4.3). The balance weight λ in Eq. (5) is set as 200 for ImageNet and 80 for CIFAR-10 due to differences between datasets and the ablation study (Section 4.3). The hyper-parameter $\tau = -0.1$ in Eq. (4) and the proportion $\phi = 10\%$ in Eq. (5) are empirical values.

4.2. Comparison to other methods

4.2.1. Results under single source model setup

In the single source model setup, only **one** source model is available for attack methods. Under this setup, we test each at-

¹ We use pre-trained ImageNet models provided by Pytorch and pre-trained CIFAR-10 models from https://github.com/Att100/CIFAR10_Pytorch.

Table 1

Transferability (%) on ImageNet (mean $\pm$ std). Source model: ResNet50. Target models are shown in the first row. **Bold** and underlined denote the highest and the second highest value in each column.

Methods	ResNet18	VGG16	DenseNet121	GoogleNet	MobileNetV2	InceptionV3	SqueezeNet	AlexNet
DeepFool (CVPR 16)	28.08 $\pm$ 2.14	25.65 $\pm$ 1.67	22.50 $\pm$ 1.64	27.26 $\pm$ 1.44	25.39 $\pm$ 1.83	28.18 $\pm$ 1.43	39.24 $\pm$ 2.45	40.76 $\pm$ 2.37
SparseFool (CVPR 19)	39.93 $\pm$ 1.52	39.95 $\pm$ 1.80	30.82 $\pm$ 1.15	39.10 $\pm$ 1.42	34.02 $\pm$ 1.19	38.68 $\pm$ 1.70	57.61 $\pm$ 1.46	52.61 $\pm$ 2.21
I-FGSM (ICLR W 17)	29.26 $\pm$ 1.61	26.93 $\pm$ 1.31	24.24 $\pm$ 1.50	28.60 $\pm$ 0.90	27.01 $\pm$ 1.71	29.37 $\pm$ 1.35	40.25 $\pm$ 2.20	41.11 $\pm$ 2.08
NI-FGSM (ICLR 20)	49.67 $\pm$ 1.22	44.00 $\pm$ 1.44	47.61 $\pm$ 1.55	39.81 $\pm$ 1.47	45.73 $\pm$ 1.85	37.87 $\pm$ 0.91	51.19 $\pm$ 1.99	44.81 $\pm$ 2.02
MI-FGSM (CVPR 18)	49.76 $\pm$ 1.23	44.00 $\pm$ 1.20	47.56 $\pm$ 1.47	39.65 $\pm$ 1.69	45.64 $\pm$ 1.88	37.72 $\pm$ 0.89	51.35 $\pm$ 1.98	44.79 $\pm$ 1.99
SmoothFool-U ² (WACV 20)	27.90 $\pm$ 1.38	25.95 $\pm$ 1.89	23.76 $\pm$ 1.29	28.53 $\pm$ 1.53	26.53 $\pm$ 1.49	29.08 $\pm$ 1.14	40.03 $\pm$ 1.93	42.11 $\pm$ 1.77
SmoothFool-L ² (WACV 20)	29.21 $\pm$ 1.09	26.08 $\pm$ 1.83	24.59 $\pm$ 1.35	28.60 $\pm$ 1.49	27.09 $\pm$ 1.36	29.05 $\pm$ 1.33	40.46 $\pm$ 1.93	42.76 $\pm$ 1.86
SmoothFool-G ² (WACV 20)	29.31 $\pm$ 1.12	26.14 $\pm$ 1.80	24.73 $\pm$ 1.32	28.63 $\pm$ 1.38	27.12 $\pm$ 1.42	29.16 $\pm$ 1.32	40.46 $\pm$ 2.00	42.82 $\pm$ 1.85
ColorAdv (NeurIPS 19)	<u>52.59$\pm$2.09</u>	<u>50.77$\pm$1.37</u>	<u>49.38$\pm$1.74</u>	<u>43.77$\pm$1.47</u>	<u>51.98$\pm$1.61</u>	<u>42.22$\pm$1.30</u>	<u>59.13$\pm$1.59</u>	<u>56.74$\pm$2.08</u>
GMM (Ours)	62.98$\pm$1.18	60.07$\pm$1.05	53.16$\pm$1.44	53.97$\pm$2.11	59.04$\pm$1.65	54.10$\pm$1.55	75.61$\pm$1.16	78.85$\pm$1.19

tack method on three source models: ResNet50, VGG16, and InceptionV3, respectively.

Transferability Evaluation. Table 1 shows the transferability of adversarial examples generated by different methods on 10 sets of ImageNet images where the source model is ResNet50. The results are reported in the form “mean $\pm$ std”.

From Table 1, GMM significantly increases the transferability to above 50% on all target models. Moreover, our method achieves 5–20% higher transferability than ColorAdv, which is the best in compared methods.

In terms of other transfer-based methods, MI-FGSM and NI-FGSM improve the optimization algorithm based on I-FGSM in different ways and boost the transferability by 5–20%. But their final results are below 50% under most target models which implies that refining optimization algorithms can improve limited transferability. SmoothFool² also gets limited improvement of transferability as Table 1 shows. This result may reflect that local smooth perturbations do not help transferability that much.

From Table 1, the transferability of adversarial examples generated by all methods varies for different target models. However, the transferability of adversarial examples generated by GMM varies much more than other attacks. There is an observation that, compared with ColorAdv, GMM improves the transferability by about 10% when the target model is ResNet18, VGG16, DenseNet121, GoogleNet, MobileNetV2 or InceptionV3, but improves the transferability by about 20% when the target model is SqueezeNet or AlexNet. Much higher improvement on these two target models causes the transferability of adversarial examples generated by GMM varies in a larger range than other attack methods.

To investigate why adversarial examples generated by GMM transfer much better on SqueezeNet and AlexNet than other target models, we visualize some adversarial examples generated by GMM against ResNet50 which can transfer to SqueezeNet and AlexNet successfully, but fail to transfer to other target models. We find the common characteristic for such adversarial examples is, they are generated early in Algorithm 1, which means they are generated with small number of iterations. Adversarial examples with fewer iterations have relatively small perturbations added and the perturbations are even hard to be detected by humans (see Fig. 3).

There are two reasons for such adversarial examples successfully transferring to SqueezeNet and AlexNet but failing to transfer to other target models. On the one hand, SqueezeNet and AlexNet are less robust than other target models so it provides the possibility of adversarial examples with small-magnitude perturbations to transfer to these models. SqueezeNet and AlexNet are two shallow

networks whose depth is 10-layers and 8-layer respectively, while other target networks' depth varies from 16-layers to 121-layers. On the other hand, low-frequency perturbations help much on adversarial examples with small-magnitude perturbations generated by GMM transferring to SqueezeNet and AlexNet. It is observed that adversarial examples with small-magnitude but high-frequency perturbations³ still cannot transfer to less robust target models (SqueezeNet and AlexNet). To sum up, it is possible for adversarial examples with small-magnitude perturbations to transfer to less robust target models (SqueezeNet and AlexNet) if the perturbations are low-frequency, but successfully transferring is still hard if perturbations are high-frequency. This phenomenon suggests that low-frequency perturbations help to improve the transferability.

Table 2 shows the transferability of adversarial examples generated by different methods on the CIFAR-10 test set where the source model is ResNet50. The results are reported in the form “mean $\pm$ std”. Similar to the findings above, GMM generates adversarial examples whose transferability is 10–20% higher than other attack methods. Moreover, the transferability results of our method are above 60% on all target models.

Transferability experiments on the other two source models, VGG16 and InceptionV3, are summarized in the supplementary material. GMM also achieves the best transferability under these two models.

Attack Rate. Table 3 shows the attack rates of different methods on two datasets, while attacking ResNet50, VGG16 and InceptionV3 respectively. Although GMM does not achieve the highest attack rate for all models, it achieves a comparable attack rate (above 90%) with other attacks.

Imperceptibility of Perturbations. The imperceptibility is measured by popular IQA metrics: SSIM (Wang et al., 2004), LPIPS (Zhang et al., 2018) and PieAPP (Prashnani et al., 2018). Large SSIM, small LPIPS, and small PieAPP represent good imperceptibility. Among these metrics, SSIM is an early metric in IQA while LPIPS and PieAPP are two recent metrics based on deep learning. The latter two correlate with the human vision system better than SSIM (Prashnani et al., 2018; Zhang et al., 2018). Since GMM uses LPIPS as a part of the loss function, it is a bit unfair to compare GMM with other attacks under this metric. Therefore, PieAPP may be more appropriate for evaluating the imperceptibility of perturbations. Despite some drawbacks, SSIM and LPIPS are still chosen as two alternative metrics to quantitatively evaluate the imperceptibility of perturbations.

² In Table 1–4, SmoothFool-U/L/G represents SmoothFool with uniform kernels, linear kernels and Gaussian kernels, respectively.

³ Here, we mainly focus on adversarial examples generated by DeepFool and ColorAdv because: (1) both of these methods generate adversarial examples with high-frequency perturbations, (2) perturbations generated by DeepFool are recognized to have the smallest ℓ_2 norm, (3) ColorAdv has the best transferability in compared methods.



Fig. 3. Some adversarial examples generated by GMM with ResNet50 as the source model (zoom in for a better view). These adversarial examples can transfer to SqueezeNet and AlexNet successfully, but fail to transfer to other target models.

Table 2

Transferability (%) on CIFAR-10 (mean±std). Source model: ResNet50. Target models are shown in the first row. **Bold** and underlined denote the highest and the second highest value in each column.

Methods	ResNet18	VGG16	DenseNet121	GoogleNet	MobileNetV2	InceptionV3	SqueezeNet	AlexNet
DeepFool (CVPR 16)	50.51±1.55	47.92±1.13	48.78±1.57	53.64±1.32	51.29±1.47	48.25±1.24	49.79±1.70	45.24±1.24
SparseFool (CVPR 19)	<u>64.65±1.38</u>	<u>58.38±1.21</u>	<u>58.88±1.31</u>	<u>61.36±0.96</u>	47.75±0.90	50.40±1.78	58.06±1.94	<u>61.99±1.28</u>
I-FGSM (ICLR W 17)	26.82±1.98	21.96±1.93	21.93±1.50	30.39±2.22	22.31±2.85	22.17±1.89	32.70±2.16	36.49±2.27
NI-FGSM (ICLR 20)	57.56±2.80	52.74±1.62	56.60±1.74	56.47±1.90	<u>57.92±1.96</u>	<u>53.18±1.61</u>	<u>51.41±2.32</u>	46.80±2.30
MI-FGSM (CVPR 18)	57.59±2.81	52.67±1.65	56.53±1.85	56.48±1.80	57.91±1.97	53.14±1.68	51.39±2.35	46.75±2.31
SmoothFool-U (WACV 20)	27.89±1.50	24.78±1.46	26.22±1.45	34.92±1.49	26.15±1.43	28.13±0.86	32.85±1.84	33.37±1.62
SmoothFool-L (WACV 20)	33.31±1.27	21.34±1.53	23.28±1.20	29.29±1.57	20.90±1.56	19.30±1.02	42.97±1.45	52.17±1.60
SmoothFool-G (WACV 20)	37.31±0.92	23.69±1.44	26.06±0.85	30.81±1.34	22.63±1.53	20.19±1.31	46.13±1.60	57.85±1.65
ColorAdv (NeurIPS 19)	48.06±1.98	41.12±1.96	46.00±1.39	49.41±2.15	40.28±2.43	38.73±2.62	45.78±2.39	49.06±2.02
GMM (Ours)	73.55±1.19	68.98±1.63	73.58±1.03	67.84±1.92	67.00±1.39	67.53±1.30	70.59±1.76	72.59±1.22

Table 3

Attack Rate (%) on ImageNet and CIFAR-10. **Bold** and underlined denote the highest and the second highest value in each column.

Methods	ImageNet			CIFAR-10		
	ResNet50	VGG16	InceptionV3	ResNet50	VGG16	InceptionV3
DeepFool (CVPR 16)	91.12±0.82	89.67±0.71	89.63±1.04	92.22±0.83	91.68±0.93	<u>92.10±0.78</u>
SparseFool (CVPR 19)	90.82±1.04	90.55±1.18	89.90±0.87	<u>92.82±0.81</u>	<u>92.52±0.93</u>	92.81±0.89
I-FGSM (ICLR W 17)	91.52±0.74	89.86±0.97	87.34±0.69	66.93±1.50	71.42±1.44	69.32±1.17
NI-FGSM (ICLR 20)	93.26±0.72	93.19±0.50	87.80±0.68	62.15±1.69	66.85±1.97	62.40±1.44
MI-FGSM (CVPR 18)	93.26±0.71	93.22±0.49	87.81±0.66	62.18±1.70	66.85±1.97	62.38±1.49
SmoothFool-U (WACV 20)	90.77±0.59	89.74±1.26	89.18±0.99	92.23±0.82	91.85±0.86	92.04±0.79
SmoothFool-L (WACV 20)	90.77±0.59	89.75±1.27	89.17±0.99	92.10±0.70	91.73±0.99	91.90±0.86
SmoothFool-G (WACV 20)	90.79±0.57	89.75±1.26	89.17±0.99	92.08±0.64	91.74±1.04	91.94±0.80
ColorAdv (NeurIPS 19)	94.35±0.81	94.66±0.59	<u>94.50±0.61</u>	70.87±1.92	74.46±1.19	66.49±1.60
GMM (Ours)	<u>93.27±0.93</u>	<u>93.76±0.57</u>	96.98±0.72	93.20±0.69	93.26±0.71	91.29±0.71

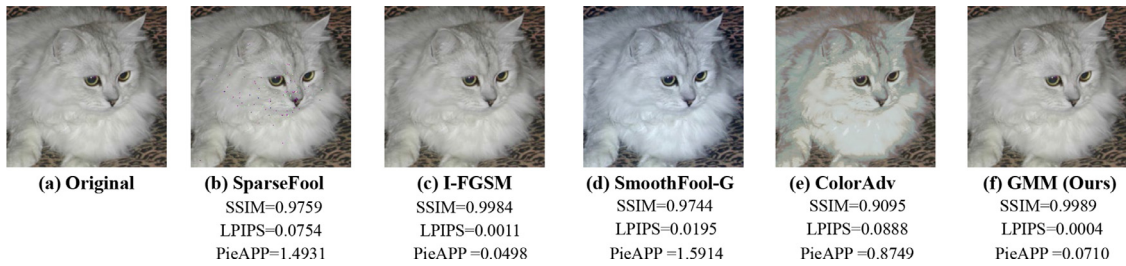


Fig. 4. Adversarial examples with corresponding attack methods and three IQA metrics.

We give some visualization results of adversarial examples generated by different methods in Fig. 4 with quantitative evaluations labeled below. More visualization results for all attacks are shown in the supplementary material. From Fig. 4, adversarial examples generated by GMM and I-FGSM look almost the same as the original image, while for other attacks, we can distinguish some differences between adversarial examples and the original image.

Overall Performance. Table 4 shows the overall performance of all compared methods under all evaluation metrics when the source model is ResNet50. The metrics are reported on 10 sets of 1000 ImageNet images in the “mean±std” form. Due to the space limit, only partial target models are shown in Table 4. The full overall performance can be found in the supplementary material. From Table 4, GMM achieves the highest transferability. Mean-

Table 4

Overall performance on 10 sets of ImageNet images. The transferability is calculated with the source model ResNet50 and the corresponding target model. **Bold** and underlined denote the highest and the second highest value in each column.

Method	Attack Rate (%)			Imperceptibility			Transferability (%)		
	ResNet50	SSIM $\uparrow$	LPIPS $\downarrow$	PieAPP $\downarrow$	ResNet18	VGG16	InceptionV3		
DeepFool (CVPR 16)	91.12 $\pm$ 0.82	0.998$\pm$0.000	0.0006$\pm$0.0001	0.036$\pm$0.001	28.08 $\pm$ 2.14	25.65 $\pm$ 1.67	28.18 $\pm$ 1.43		
SparseFool (CVPR 19)	90.28 $\pm$ 1.04	0.974 $\pm$ 0.001	0.0701 $\pm$ 0.0017	1.278 $\pm$ 0.012	39.93 $\pm$ 1.52	39.95 $\pm$ 1.80	38.68 $\pm$ 1.70		
I-FGSM (ICLR W 17)	91.52 $\pm$ 0.74	<u>0.997$\pm$0.000</u>	<u>0.0007$\pm$0.0002</u>	<u>0.037$\pm$0.001</u>	29.26 $\pm$ 1.61	26.93 $\pm$ 1.31	29.37 $\pm$ 1.35		
NI-FGSM (ICLR 20)	93.26 $\pm$ 0.72	0.979 $\pm$ 0.000	0.0158 $\pm$ 0.0005	0.210 $\pm$ 0.003	49.67 $\pm$ 1.22	44.00 $\pm$ 1.44	37.87 $\pm$ 0.91		
MI-FGSM (CVPR 18)	93.26 $\pm$ 0.71	0.978 $\pm$ 0.000	0.0157 $\pm$ 0.0004	0.210 $\pm$ 0.002	49.76 $\pm$ 1.23	44.00 $\pm$ 1.20	37.72 $\pm$ 0.89		
SmoothFool-U (WACV 20)	90.77 $\pm$ 0.59	0.852 $\pm$ 0.007	0.2305 $\pm$ 0.0102	2.237 $\pm$ 0.048	27.90 $\pm$ 1.38	25.95 $\pm$ 1.89	29.08 $\pm$ 1.14		
SmoothFool-L (WACV 20)	90.77 $\pm$ 0.59	0.968 $\pm$ 0.001	0.0255 $\pm$ 0.0004	1.195 $\pm$ 0.015	29.21 $\pm$ 1.09	26.08 $\pm$ 1.83	29.05 $\pm$ 1.33		
SmoothFool-G (WACV 20)	90.79 $\pm$ 0.57	0.967 $\pm$ 0.001	0.0259 $\pm$ 0.0003	1.196 $\pm$ 0.016	29.31 $\pm$ 1.12	26.14 $\pm$ 1.80	29.16 $\pm$ 1.32		
ColorAdv (NeurIPS 19)	94.35$\pm$0.81	0.938 $\pm$ 0.001	0.0494 $\pm$ 0.0010	0.605 $\pm$ 0.008	<u>52.29$\pm$2.09</u>	<u>50.77$\pm$1.37</u>	<u>42.22$\pm$1.30</u>		
GMM (Ours)	<u>93.27$\pm$0.93</u>	0.950 $\pm$ 0.002	0.0197 $\pm$ 0.0015	0.385 $\pm$ 0.025	62.98$\pm$1.18	60.07$\pm$1.05	54.10$\pm$1.55		

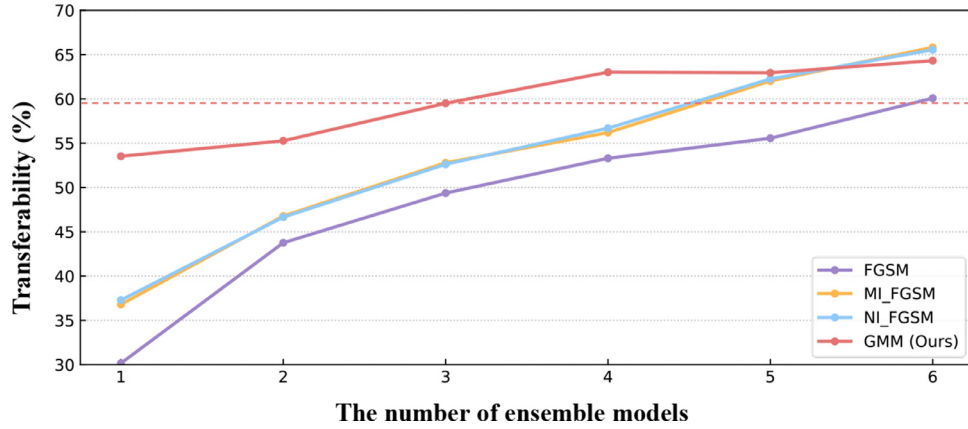


Fig. 5. Transferability evaluation of different methods under the ensemble source models setup on 1000 ImageNet images.

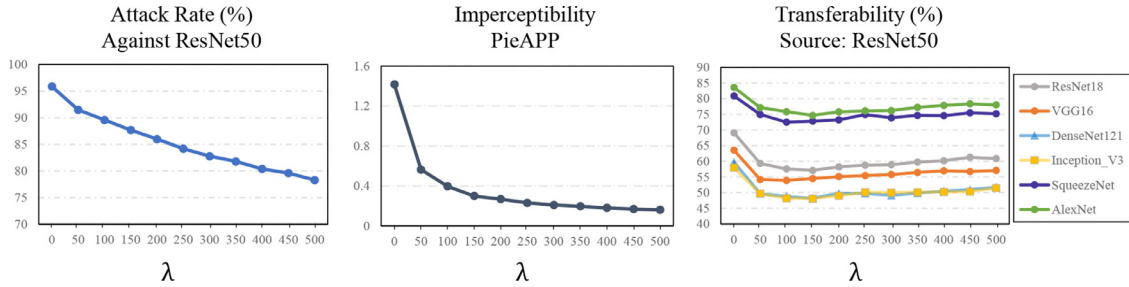


Fig. 6. Ablation study of λ . Since SSIM, LPIPS, PieAPP show similar trends, we only draw PieAPP for the simplicity of the figure.

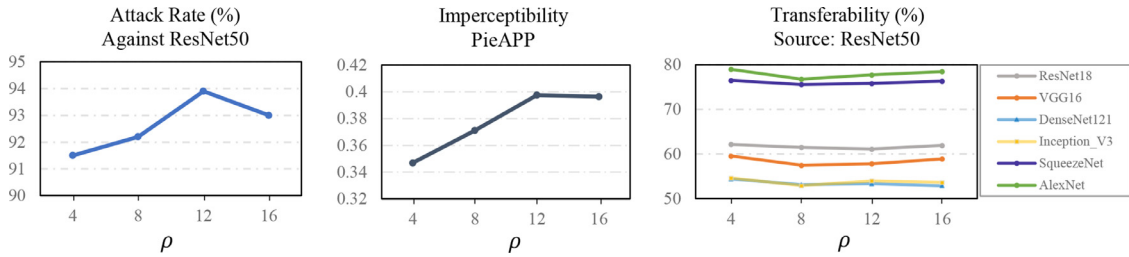


Fig. 7. Ablation study of ρ . Since SSIM, LPIPS, PieAPP show similar trends, we only draw PieAPP for the simplicity of the figure.

while, GMM maintains a comparable attack rate and perturbations' imperceptibility with other attacks.

4.2.2. Results under ensemble source models setup

The ensemble source models setup was first proposed by Dong et al. (2018) where **several** source models are available for attack methods. In detail, suppose the available source models are $M_{s_1}, M_{s_2}, \dots, M_{s_N}$ and the adversarial loss for M_{s_k} ($k = 1, \dots, N$) is

$\mathcal{L}_{adv}(\mathbf{I}, M_{s_k})$, then attackers need to minimize the ensemble adversarial loss

$$\mathcal{L}_{adv}^{ens} = \sum_{k=1}^N w_k \mathcal{L}_{adv}(\mathbf{I}, M_{s_k}) \quad (7)$$

where w_k is the weight for M_{s_k} (Dong et al., 2018; Lin et al., 2020).

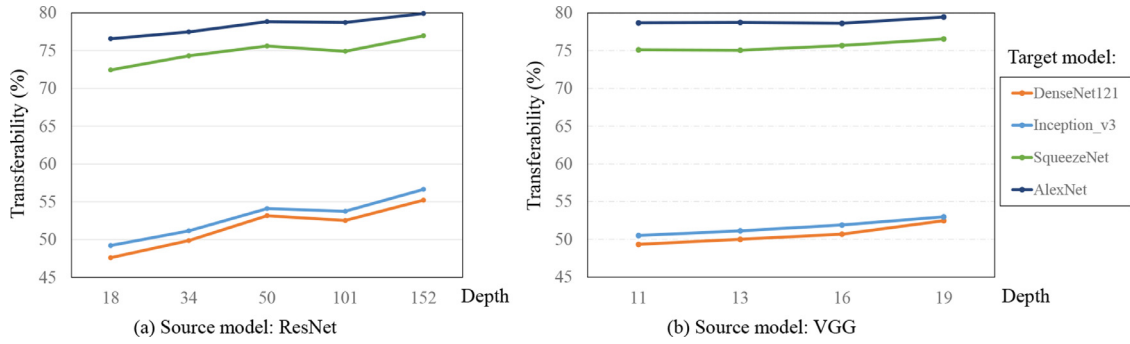


Fig. 8. The dependency of transferability from the depth of source models. ResNet and VGG with different depth are chosen as source models. The depth of the source model is shown at the horizontal axis. The vertical axis represents the mean transferability from the source mode to target models over 10 sets of ImageNet images. Lines with different colors illustrate different target models.

Table 5

Transferability (%) (the source model is ResNet50) and attack rate (%) on 1000 ImageNet images with different K for GMM.

Metric	Model	4	8	12	16	20	24	28
Transferability (%)	ResNet18	70.1	67.7	64.1	63.1	63.0	60.5	59.8
	VGG16	65.4	63.4	59.3	58.4	61.2	57.1	57.5
	InceptionV3	58.7	56.8	55.0	54.4	53.5	52.9	51.8
Attack Rate (%)	ResNet50	74.9	84.4	88.6	90.3	93.4	94.1	94.8

As for GMM, the adaptive loss in this setup is

$$\mathcal{L}^{\text{ens}} = \begin{cases} \sum_k w_k \mathcal{L}_{\text{adv}}(\mathbf{I}, M_{s_k}) & |f > J|_{\%} < \phi, \\ \sum_k w_k \mathcal{L}_{\text{adv}}(\mathbf{I}, M_{s_k}) + \lambda \cdot \mathcal{L}_{\text{dis}} & \text{otherwise} \end{cases} \quad (8)$$

where $\mathcal{L}_{\text{adv}}$ is formulated as Eq. (4).

Following the work (Lin et al., 2020), we compare GMM with FGSM, MI-FGSM, and NI-FGSM in the ensemble source models setup and assign equal weights for different source models, i.e. $w_k = 1/N$ ($k = 1, \dots, N$) where N is the number of source models.

We incrementally increase the number of source models from 1 to 6 in the order: ResNet50, VGG16, ResNet18, DenseNet121, SqueezeNet, and AlexNet. The target model is InceptionV3. Adversarial examples' transferability of different methods is compared under one source model, two source models, ..., six source models, respectively.

Transferability Evaluation. Fig. 5 shows the transferability on 1000 ImageNet images when we incrementally add source models. GMM achieves nearly 60% transferability with 3 source models while other methods need at least 5 models to achieve comparable transferability. This result means GMM is efficient in generating transferable adversarial examples because it can use less source models to achieve comparable or even better transferability with other attacks.

4.3. Ablation studies

In this subsection, ablation studies are conducted on ImageNet because it is more challenging than CIFAR-10. Experiments are applied on 1000 images randomly chosen from ImageNet, belonging to 1000 classes. Moreover, the setup is the single source model setup.

The number of Gaussian bases. To study how the number of Gaussian bases K in Eq. (2) affects attack rate and transferability, we evaluate GMM with $K = 4, 8, \dots, 28$. The attack rate on ResNet50 and the transferability on ResNet18, VGG16, InceptionV3 is shown in Table 5.

From Table 5, the choice of K is a trade-off between the attack rate and the transferability. The increase of K leads to more diverse and complex perturbations and thereby increases the attack rate. At the same time, larger K also means more parameters,

which results in overfitting of perturbations to the source model. Hence, larger K results in lower transferability. We choose $K = 20$ to reconcile the attack rate and the transferability.

The parameterized function. In addition to the GMM, other parameterized representations of functions can also be applied to the proposed method. We test our method using the following three parameterized representations with different degrees in Eq. (1), including: (i) linear polynomial: $f(x, y; \Theta) = \theta_1 + \theta_2 x + \theta_3 y$; (ii) quadratic polynomial: $f(x, y; \Theta) = \theta_1 x^2 + \theta_2 y^2 + \theta_3 xy$; (iii) linear trigonometric polynomial: $f(x, y; \Theta) = \theta_1 \sin(x) + \theta_2 \sin(y) + \theta_3$.

We compare their performance with GMM and report the adversarial examples' transferability and the attack rates in Table 6. Surprisingly, adversarial examples' transferability with simple smooth functions is much higher than that with GMM. However, their attack rates are too low to be satisfactory.

This phenomenon may reflect that the number of parameters in f can make a trade-off between the attack rate and transferability. On one hand, simple functions have fewer parameters than GMM, so they prevent perturbations from overfitting the source model better than GMM. On the other hand, their number of parameters is too small to generate various perturbations and attack the source model successfully.

The balance weight λ between $\mathcal{L}_{\text{adv}}$ and $\mathcal{L}_{\text{dis}}$. Intuitively speaking, the larger λ is, the more important $\mathcal{L}_{\text{dis}}$ is. Hence, the generated adversarial example will be more similar to the original image. Fig. 6 shows the ablation study of λ in Eq. (5) when the source model is ResNet50. As λ increases, the attack rate goes down but the imperceptibility of perturbations gets better. The result is in line with our expectations. We also find that as long as λ is not zero, it has little effect on transferability.

The perturbation scale ρ . Fig. 7 shows the ablation study of ρ in Eq. (1). As ρ increases, the attack rate increases slightly while the imperceptibility of perturbations gets worse. The phenomenon is because that large perturbations are added to input images as ρ increases. Therefore, it is easier to attack models, but the perturbations become more visible instead. Overall, ρ has limited impact on transferability.

The depth of the source model. This experiment is about the dependency of transferability from the depth of source models.

Table 6

Transferability (%) (the source model is ResNet50) and attack rate (%) of different parameterized functions on 1000 ImageNet images. **Bold** and underlined denote the highest and the second highest value in each column.

Evaluations Functions	Transferability (%)			Attack Rate (%)		
	ResNet18	VGG16	InceptionV3	ResNet50	VGG16	InceptionV3
Linear	87.63	83.87	<u>74.73</u>	37.2	43.4	40.7
Quadratic	<u>86.92</u>	<u>83.37</u>	75.75	36.7	43.9	41.4
Trigonometric	86.47	82.71	74.19	<u>39.9</u>	<u>45.8</u>	<u>44.5</u>
GMM	62.96	61.24	53.53	93.4	92.9	96.9

It is conducted under the single source model setup. We choose two classes of models as source models where models in the same class have the same architecture but different depths: (1) ResNet: ResNet18, ResNet34, ResNet50, ResNet101, ResNet152; (2) VGG: VGG11, VGG13, VGG16, VGG19. These networks are chosen because Pytorch provides their pre-trained models on ImageNet.

We report the mean transferability of adversarial examples generated by GMM over 10 sets of ImageNet images from ResNet or VGG to some chosen target models in Fig. 8. The transferability improves as the source model goes deeper. This may be because deep networks can extract more task-relevant features compared with shallow networks. Therefore, adversarial examples generated at the output layer of deep networks may be more task-relevant and may transfer better to other models.

Discussion. Ablation studies on K and parameterized functions help us determine the parameters of GMM in our work. Ablation studies on λ and ρ show that they have little impact on transferability. This phenomenon illustrates that the transferability improvement of GMM does not depend on these adjustable hyperparameters. Instead, the improvement depends on the design of parameterized functions and global smooth perturbations.

5. Conclusion and future work

In this paper, we propose the GMM attack, a new way to generate global smooth perturbation by parameterized smooth functions. This class of perturbations can significantly increase adversarial examples' transferability due to its smoothness and low-frequency characteristics. Moreover, attackers can generate diverse low-frequency perturbations with different functions (even simple functions such as linear functions can generate transferable adversarial examples). We conduct extensive experiments to show the superiority of our attack method. For future work, we intend to consider more parameterized smooth functions and to investigate theoretical explanations for the high transferability of the proposed class of perturbations.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Yujia Liu: Conceptualization, Methodology, Investigation, Writing – original draft, Visualization. **Ming Jiang:** Conceptualization. **Tingting Jiang:** Conceptualization, Methodology, Writing – review & editing, Resources.

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